

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/313240833>

Application of rapid prototyping in maxillofacial surgery

Conference Paper · January 2016

CITATIONS

5

READS

14,299

5 authors, including:



Aleksandar Dimic

University of Belgrade

8 PUBLICATIONS 43 CITATIONS

SEE PROFILE



Zarko Miskovic

University of Belgrade - Faculty of Mechanical Engineering

68 PUBLICATIONS 340 CITATIONS

SEE PROFILE



Drago B Jelovac

University of Belgrade

65 PUBLICATIONS 715 CITATIONS

SEE PROFILE



Radivoje Mitrovic

University of Belgrade

102 PUBLICATIONS 697 CITATIONS

SEE PROFILE

Application of rapid prototyping in maxillofacial surgery

Aleksandar DIMIC
Zarko MISKOVIC
Drago JELOVAC
Radivoje MITROVIC
Mileta RISTIVOJEVIC
Marija MAJSTOROVIC

Abstract: Nowadays, despite very high precision of medical procedures and equipment, improving of pre-operative and operative processes is always desirable. This multidisciplinary paper deals with the application of 3D modelling and printing in the pre-operational planning stages in maxillofacial surgery. By applying of these methods significant benefits were achieved before and during the operation, the time patient spent in anaesthesia is shortened, and the potential for error is significantly reduced, which is the main goal of all trends in modern medical science.

Key words: rapid prototyping, 3D printing, modeling, maxillofacial surgery

1. INTRODUCTION

Taking into account most recent discoveries, 21st century medicine, with all related fields, is definitely advancing faster than in 20th century. Significant results have been achieved in the field of internal medicine, orthopaedics, oral surgery, prosthodontics, surgery, etc. Maxillofacial surgery is definitely not left behind. One of the most significant developments arises from the possibility of replacing the damaged bone of the maxillofacial region with another bone from different region or some biocompatible implant. In order to achieve high accuracy of these methods it was necessary to develop a sufficiently precise algorithms, apparatus and technology. Improvement of modern hardware and software for human body parts scanning and the compatibility of this software with the modelling and 3D printing software enabled the connection between mechanical engineers and maxillofacial surgeons. Multidisciplinary approach in this case is of great importance because it encourages the development of new research in each of the mentioned scientific fields. The main contribution of presented

research was reflected in the preparation of plastic prototypes of damaged bones in the maxillofacial region as well as healthy bones necessary for the replacement of the foregoing. It was latter shown that the existence of these prototypes significantly contributed to the advancement of the process of implantation and reconstruction of facial bones.

2. IMPLEMENTATION OF THE PROJECT

The idea to initiate this project occurred as a result of long-lasting cooperation between the Faculty of Dental Medicine and Faculty of Mechanical Engineering at University of Belgrade. After a few constructive sessions the plan of the project was developed, and latter implemented in several stages, as shown in following chapters – through several successful cases.

2.1. Obtaining of X-ray scans

The first step in the realization of the project was to obtain an X-ray scan of the patient. After a detailed analysis of the scan and consultations with maxillofacial surgeons (who provided the scan), DICOM file was imported into the standardized software, used to generate appropriate model. A scan set, required for the further preparation, is shown in Figure 1 (red circle indicates the position of the damage hole on the lower jaw due to the effects of aggressive tumor).

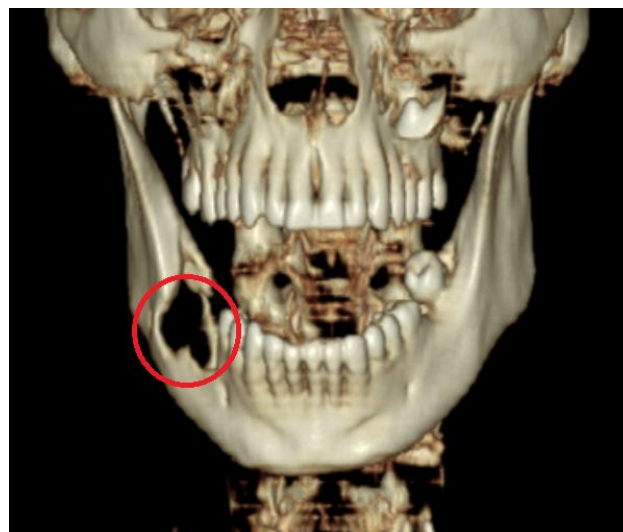


Fig 1. Preview of a head scan with damaged lower jaw bone

2.2. Scan – based prototyping

First step of the project implementation was followed by the drafting of the head model and extracting of an area of particular interest (the area of damage). During the model creation phase, the above-mentioned software provided the ability to isolate bones, skin, tissues, muscles, etc. – and as a result a shell model was obtained. The quality of scanner has a huge impact on the quality of the model. The modern multislice scanners operate on the principle of cross-sections scanning, at a relatively small distance. When a series of scanned layers, obtained in this manner,

are sequenced, the 3D image of the observed object is generated. The basic principle of this kind of scanning method is shown on Figure 2.

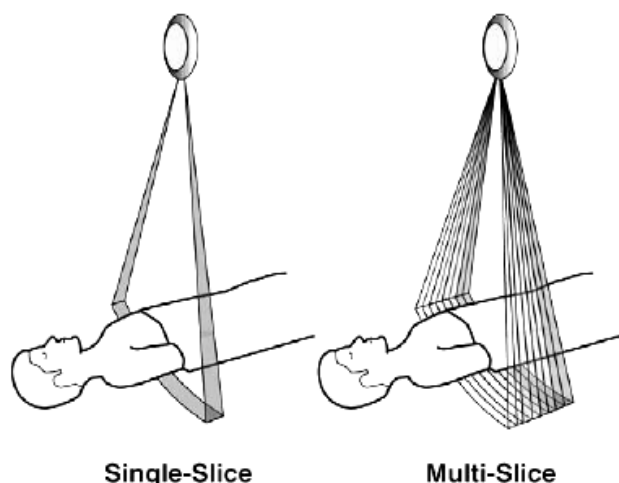


Fig 2. The difference between Single – slice and Multi – slice scanning methods

When the high resolution scanners are used, the distance between two adjacent scanned slices is very small (for example 0,5mm). This can be a problem because, due to the high resolution, complete 3D model is generated from an extremely large number of slices. For processing of such models, a computer with exceptional capacities is needed. The opposite extremes are scanners with very low resolution which, due to the large distance between the two slices, can cause the loss of important information which are located between these slices. The Figure 3 shows a model obtained from a head scan when only bones are isolated. The damage of the lower jaw can be clearly seen.

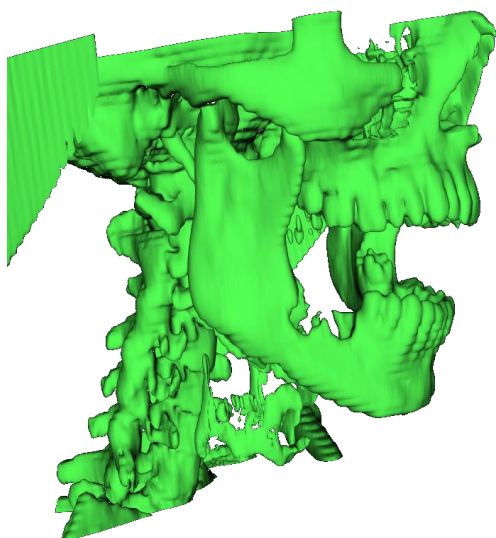


Fig 3. Shell model of the damaged jaw bone

2.3. Making of the printable 3D model

Since the standardized 3D recognition software enables only the extraction of the shell model (surface model with

infinitely thin external walls) in order to obtain printable model further processing is necessary. The main reason for this is the fact that rapid prototyping software does not recognize the surface models and therefore printing phase can't be accessed. At the moment, there are several methods for the conversion of a shell model into a finite volume model. It was decided that this action should be accomplished by adding of a certain amount of material, on the inside surface of the outer wall (shell). Considering that the material is added only to the inside surface of the wall, external parties, especially geometry of damaged bone, remains unchanged. This whole procedure is performed in other standardized software, and a model, after the corrections, has the appearance shown in Figure 4.

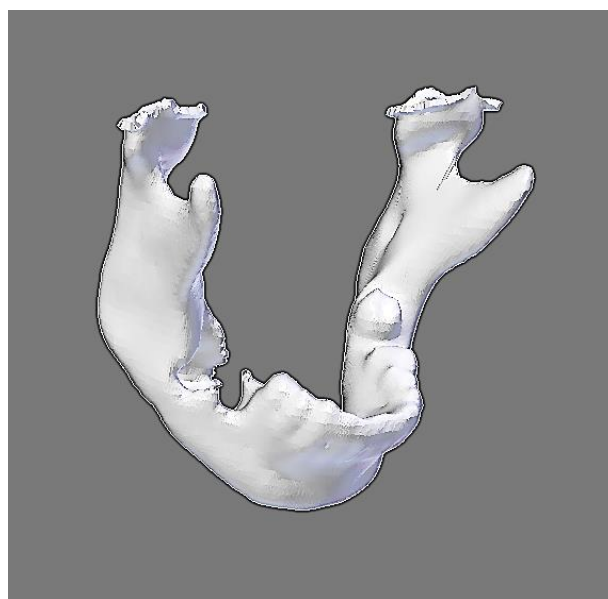


Fig 4. The appearance of volumetric model of damaged jaw bone

2.4. Additional adjusting of the printable 3D model

The final processing of obtained 3D model is performed in the third standardized software, because it has a number of very useful options, used for model simplification – significantly affecting its printing performances. Also, this program serves to remove unnecessary parts of the model and to isolate the desired parts of the scanned human body. In the presented case, the lower jaw bone was successfully extracted from the rest of the head. For that purpose, several commands were used, such as *smoothing* using the Laplacian smooth tool, or *removal of small isolated areas* of the model. These small particles of the model are either randomly generated, or are really part of the model, but they are so small in size that their elimination does not affect the quality of the final model, while on the other hand, it has a significant impact on 3D print facilitating. Work in an environment of mentioned standardized software is shown in Figure 5.

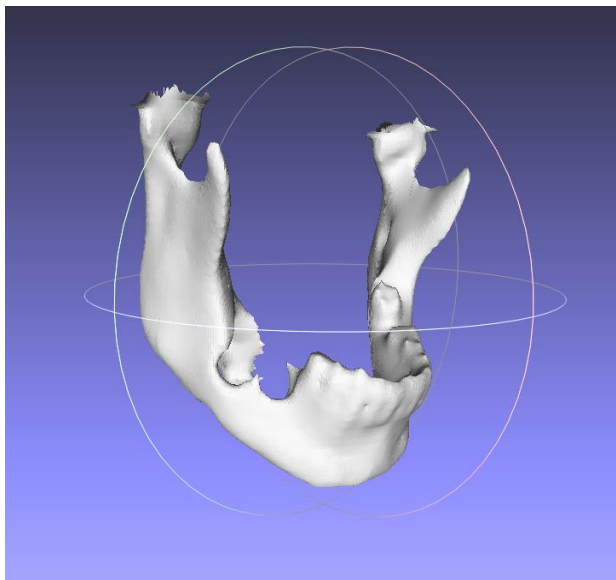


Fig 5. Finishing touches on a printable 3D model

3. 3D PRINTING OF MODEL

After the completion of all previously listed actions, generated printable 3D model was sent to the 3D printer. Printing process was very complicated because it requires the definition of the huge number of parameters whose values can't be found in guide books, so they have the exclusive experimental character. Taking this into account, every model is a story in itself, and as such requires the definition of its own printing parameters. 3D printing works on the principle of bringing the plastic thread to the extruder where it is melted down and delivered to the printing platform. The printed prototype, as well as scan, also consists of layers of melted plastic, where great care is needed as well as the constant presence of the user, because a mistake made in any printing layer entails generating and superimposing errors in all subsequent layers.

3.1. Definition of 3D printing parameters

Some of the main parameters of the 3D printing are:

Extruder temperature – the temperature of the extruder should be optimal, as it is directly related to the speed of adding of the melted plastic to the printing platform.

The speed of material adding – as already mentioned, it strictly depends on the temperature of the extruder, which melts the plastic. The combination of these two parameters must ensure adequate melting and bonding of plastic interlayer. Excessive rate of feed leads to the pouring of plastic in undesirable places. On the other hand, insufficient rate of feed leads to porosity of 3D printed model. In addition to these, another very important parameter is the temperature of plate.

Temperature of plate (printing platform) – it must ensure that there is no contact break between the model and the plate on which the model is set. The plate and printing head gradually depart during printing. This translation allows the formation of a model from a large number of layers. When a printing layer is completed, the printing

head is elevated for a specific distance (depending on desired layer thickness), and then the procedure continues. Regarding to the parameters influencing the structure of the model, the major are:

Layer thickness – this parameter is very important because it directly affects the surface quality of the model. It also indicates whether a particular piece of the model will be completed in one, two or more passes of extruder – thus affecting the overall printing time. On the other hand, a situation when the thickness of the model is smaller than defined thickness of molten plastic layer must be avoided, because, in that case, the degradation of the model geometry occurs.

Infill percent – the structural integrity and overall printing time of the specific 3D model are directly dependent of this factor, taking into account that it defines the percentage of material inside the printed model volume.

The position of the model in the printing area is also of great importance. An unfavorable position of model leads to a situation where software has to generate artificial support of the printed model (so-called supports), thereby unnecessarily increasing the printing time and risk of printing failure. Positioning of the damaged lower jaw bone 3D model in printing area is shown in the figure below (Fig. 6).

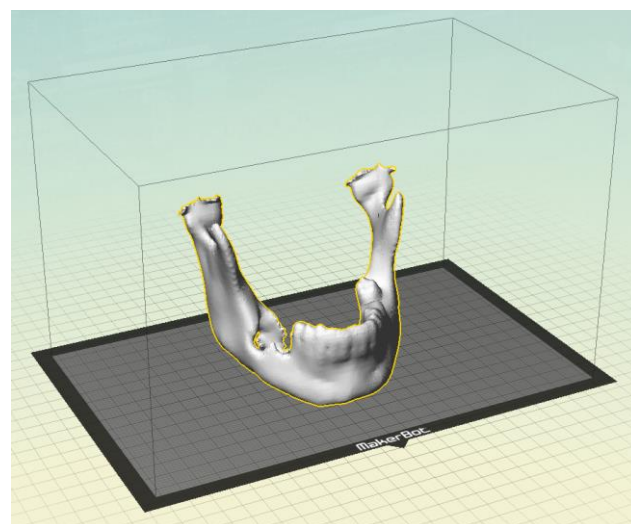


Fig 6. Positioning of the damaged lower jaw bone 3D printing model on printing platform

3.2. Printing material

As a material for 3D printing, the industrial-strength Acrylonitrile Butadiene Styrene (ABS) plastic was used. Its detailed mechanical and physical characteristics are given in the Table 1. ABS is derived from acrylonitrile, butadiene, and styrene. Acrylonitrile is a synthetic monomer produced from propylene and ammonia; butadiene is a petroleum hydrocarbon obtained from the C4 fraction of steam cracking; styrene monomer is made by dehydrogenation of ethyl — a hydrocarbon obtained in the reaction of ethylene and benzene.

ABS plastic combines the strength and rigidity of acrylonitrile and styrene polymers with the toughness of polybutadiene rubber. While the cost of ABS producing is

roughly twice the cost of producing polystyrene, it is considered superior for its hardness, gloss, toughness, and electrical insulation properties. Also, ABS plastic is easily machined (with common machining techniques, including turning, drilling, milling, sawing, die-cutting and shearing) – making it the most common plastic for industrial usage worldwide. ABS plastic can be also cut with standard shop tools and line bent with standard heat strips.

Table 1. Physical and mechanical properties of ABS plastic

ABS plastic		
Physical properties	Density	0,9-1,53 [g/cm ³]
	Thermal conductivity	0,1 [W/mK]
	Linear thermal expansion coefficient	73,8·10 ⁻⁶ [m/mK]
	Temperature range of usefulness	-20-80 [°C]
Mechanical properties	Tensile strength	22 [MPa]
	Tensile modulus	1360 [Mpa]
	Tensile elongation	6 [%]

Regarding to the displayed physical and mechanical properties of ABS plastic, it is clear why this material is most often chosen for 3D printing and subsequent mechanical processing.

4. USAGE OF 3D PRINTED PROTOTYPE IN A PRE-OPERATIONAL ACTIVITIES

When a prototype of damaged jaw bone was finally manufactured, it was mechanically trimmed, getting final shape. The model was then sent to the maxillofacial surgeons. In this particular case, damaged part of the lower jaw bone was replaced by the upper part of the hip bone. The upper part of the hip bone was chosen in this application due to the high similarity with geometry of the lower jaw (so-called U-shape), as shown on the Figure 7. The procedure of hip bone prototyping is identical as for jaw bone and it consists of the same steps – described in previous chapters. This procedure is graphically represented on Figures 8-11.



Fig 7. Matching of the geometries of the 3D printed upper part of the hip and lower jaw

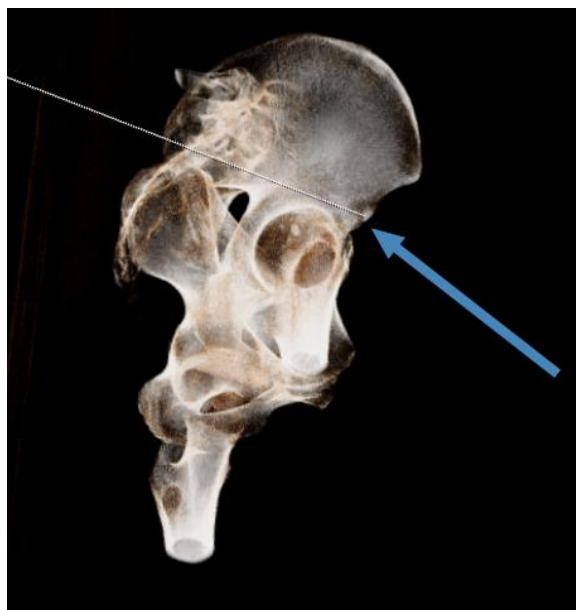


Fig 8. Preview of the upper part of hip bone scan which should be used as a implant

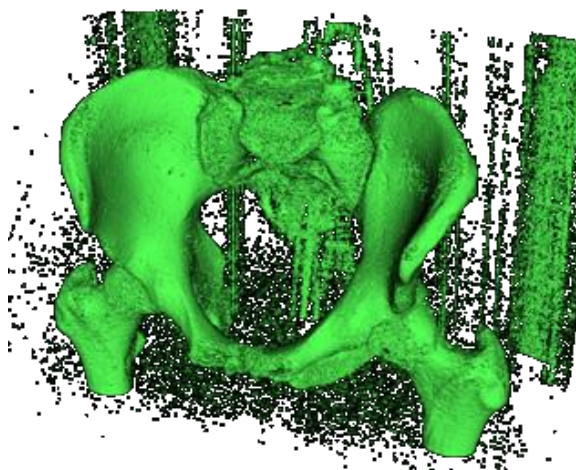


Fig 9. Shell model of hip bone

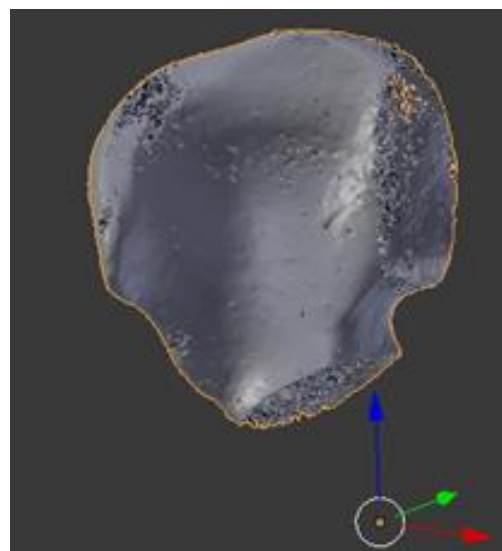


Fig 10. The appearance of volumetric model of hip bone

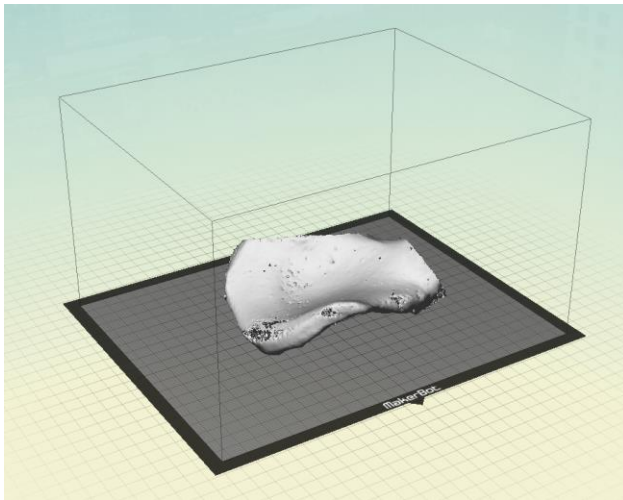


Fig 11. Positioning of the hip bone printing model in the printing area

After rapid prototyping of both presented models, surgeons had several benefits. They were able to determine exactly how foreseen healthy bone can replace the damaged one, use the model to rehearse interventions necessary to remove the damaged bone as well as those for separation of the healthy bone (implant), or determine the exact amount of material for the implementation of the operational process, etc.

For example, a small mismatch between the two shapes (red circle) can be seen on the picture shown below (Fig. 12) – without prototypes, this mismatch could be seen only after cutting of the patient. As it was observed much earlier (before the actual operation), appropriate measures could be taken pre-operatively.

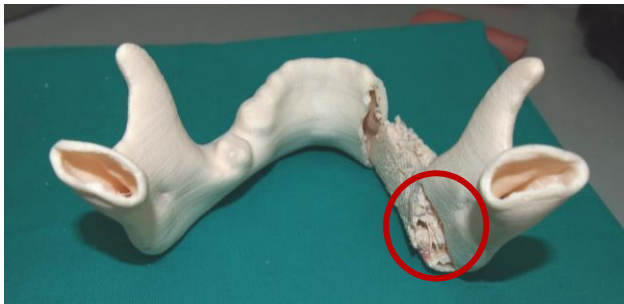


Fig 12. Spotting of the mismatch between part of hip bone (used as an implant) and damaged lower jaw bone

After the removal of the identified defects, the subsequent positioning of implant was made, as well as the shaping of the surgical guides, necessary for the fixation of the implant during and after surgery. The result of these procedures is shown on Figure 13.

The successful completion of described activities (the pre-operative process), caused the surgeons to be fully prepared for performing of successful actual operation.



Fig 13. Adaptation of the surgical guides and a matching of the printed models

5. CONCLUSION

After the completion of described multidisciplinary project, very positive feedback was obtained by the surgeons of the maxillofacial surgery. They were fully satisfied with the obtained prototypes and their fit with the real objects (prototyped human body parts). An application of this, relatively simple, technology has achieved significant effects:

- The time patient spent in the state of general anesthesia is significantly reduced because the number of operations which would have been carried out is done pre-operative.
- The precision during the execution of the operational process was significantly increased.
- Surgeons were familiarized with geometry of the damage long before the opening of the patient.
- Possible mismatches between the implant (obtained from hip bone) and the bone that is receiving the implant (lower jaw bone) can be detected and thus, the occurrence of complications during surgery was prevented pre-operative.

When all of the above is taken into consideration, it could be concluded that, by the usage of described methodology, multiple influences on the improvement of surgical procedures in the field of maxillofacial surgery could be achieved, among them – the most important: humanistic contributions and benefits of the patients.

REFERENCES

- [1] Succo G., Berrone M., Battiston B., Tos P., Goia F., Appendino P. & Crosetti E.. (2013). *Step - by - step surgical technique for mandibular reconstruction with fibular free flap: application of digital technology in virtual surgical planning*, Springer, ISSN 0937-4477, Berlin
- [2] Hidalgo DA (1989) *Fibula free flap: a new method of mandible reconstruction*. *Plast. Reconstructive Surgery* 84:71–79

- [3] Patel A, Levine J, Brecht L, Saadeh P, Hirsch DL (2012). *Digital technologies in mandibular pathology and reconstruction*. Atlas Oral Maxillofacial Surg Clin N Am 20:95–100
- [4] Foley BD, Thayer WP, Honeybrook A, McKenna S, Press S. (2013). *Mandibular reconstruction using computer-aided design and computer-aided manufacturing: an analysis of surgical results*. J Oral Maxillofac Surg 71(2):e111–e119.
- [5] Levine JP, Patel A, Saadeh PB, Hirsh DL (2012) .*Computer aided design and manufacturing in craniomaxillofacial surgery: the new state of the art*. J Craniofac Surg 23(1):288–293.
- [6] A. Mehndiratta, H. von Tengg-Kobligk, C. M. Zechmann, R. Unterhinninghofen, H.-U. Kauczor, F. L. Giesel (2010). *3D printing based on imaging data: review of medical applications*.
- [7] International Journal of Computer Assisted Radiology and Surgery
- [8] Sugar A. *3D Printing in Reconstructive Surgery*. Morriston Hospital, Swansea, Wales



Mileta RISTIVOJEVIC Prof. Ph.D.
University of Belgrade
Faculty of Mechanical engineering
Kraljice Marije 16
11000 Belgrade, Serbia
mrstivojevic@mas.bg.ac.rs



Marija MAJSTOROVIC, student
University of Belgrade
Faculty of Dental medicine
Dr. Subotica 8
11000 Belgrade, Serbia
marijavmajstorovic@gmail.com

CORRESPONDENCE



Aleksandar DIMIC, Ass.
University of Belgrade
Faculty of Mechanical engineering
Kraljice Marije 16
11000 Belgrade, Serbia
adimic@mas.bg.ac.rs



Zarko MISKOVIC, Ass.
University of Belgrade
Faculty of Mechanical engineering
Kraljice Marije 16
11000 Belgrade, Serbia
zmiskovic@mas.bg.ac.rs



Drago JELOVAC, TA. Dr Sci. Med.
University of Belgrade
Faculty of Dental medicine
Dr. Subotica 4, Department of
maxillofacial surgery
11000 Belgrade, Serbia
drago.jelovac@stomf.bg.ac.rs



Radivoje MITROVIC Prof. Ph.D.
University of Belgrade
Faculty of Mechanical engineering
Kraljice Marije 16
11000 Belgrade, Serbia
rmitrovic@mas.bg.ac.rs